Rowzatul Zannath Prerona

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Bachelor of Science
Ahsanullah University of Science and Technology,Dhaka
GitHub Profile
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Education

•Bachelor of Science in Computer Science and Engineering *2022-26*
Ahsanullah University of Science and Technology,Dhaka CGPA: 3.46

•Higher Secondary Certificate *2018-20*
Holy Cross College,Dhaka GPA: 5.00

•Secondary School Certificate *2015-18*
Ideal School and College, Motijheel, Dhaka GPA: 5.00

Academic Projects

•EfficientNet-CBAM: Attention-Guided Brain Tumor Classification
An attention-guided deep learning framework for multi-class brain tumor classification from MRI scans.

- Integrated Convolutional Block Attention Module (CBAM) into an EfficientNetB3 backbone to focus on pathologically relevant regions, achieving **99.29%** classification accuracy.
- Implemented HiResCAM gradient-based visualizations for model explainability, improving clinical interpretability across Glioma, Meningioma, and Pituitary Tumor classes.
- Technology Used: Python, TensorFlow, Keras, EfficientNetB3, CBAM, HiResCAM, OpenCV, NumPy, scikit-learn.

•Deforestation Mapping in Dhaka — Semantic Segmentation
Deep learning-based semantic segmentation to detect and monitor deforested areas using satellite imagery.

- Built a high-resolution satellite imagery dataset of Dhaka spanning 2012–2022 and benchmarked three architectures: U-Net, Attention U-Net, and a hybrid ResNet50-Attention U-Net.
- Evaluated models using Dice Score, IoU, F1-Score, Precision, and Recall; providing a reproducible framework for environmental change monitoring.
- Technology Used: Python, TensorFlow, Keras, PyTorch, U-Net, ResNet50, NumPy, OpenCV, scikit-learn, matplotlib.

•Planet Paw — Animal Rescue & Welfare Platform
A multifaceted web platform integrating donation processing, pet adoption, medical aid listings, and food sales.

- Designed a community-driven ecosystem supporting full CRUD operations for rescue listings, donor management, and inventory with emphasis on transparency and sustainability.
- Built with a PHP/MySQL backend managed via XAMPP and phpMyAdmin, delivering seamless user engagement across multiple welfare services.
- Technology Used: PHP, HTML, CSS, JavaScript, MySQL, XAMPP, phpMyAdmin.

•LeafLedger — Bookshop Management System
A full-stack bookshop management system streamlining inventory, sales, and customer operations.

- Implemented book and genre management, shopping cart functionality, stock monitoring, order processing, and admin-level operations including book approvals and content management.
- Built on ASP.NET Core MVC with a Bootstrap frontend and SQL Server database, providing a clean administrative interface suitable for real-world retail deployment.
- Technology Used: ASP.NET Core MVC, Bootstrap, HTML, CSS, SQL Server, .NET 8 SDK, SSMS.

•Spreed Love for Pet — Pet Management Desktop Application
A feature-rich Java desktop application consolidating pet management, health monitoring, and community engagement.

- Developed user and pet profile modules, an expense tracker, a pet health encyclopedia, a BMI calculator, and a community connectivity module for pet owners.
- Technology Used: Java.

•Paw Prints Manager — Pet Care Mobile Application
A comprehensive all-in-one pet care mobile application built with Flutter for pet owners.

- Features include pet and user profile management, a monthly expense tracker, a health encyclopedia searchable by concern, a BMI calculator, and a reminder system for vet visits and medications.
- Technology Used: Flutter, Dart.

Publications

- **Spectral–Spatial Mamba with Uncertainty-Guided Refinement for Thyroid Nodule Diagnosis** *2026*
  Array (Elsevier) Published
  - Co-authored with Mohammad Amanour Rahman; proposed SSMURNet, a spectral-spatial Mamba framework with evidential deep learning for confidence-calibrated thyroid nodule malignancy classification from ultrasound images, achieving 92.50% accuracy and 0.9700 AUROC.
  - DOI: 10.1016/j.array.2026.101058

- **Hard Spatial Gating for Precision-Driven Brain Metastasis Segmentation: Addressing the Over-Segmentation Paradox in Deep Attention Networks** *2025*
  arXiv preprint, eess.IV / cs.CV Preprint
  - Sole-authored work introducing SG-Net, a precision-first spatial gating architecture that resolves the "over-segmentation paradox" in brain metastasis MRI segmentation, achieving a threefold improvement in boundary precision over Attention U-Net with 8.8× fewer parameters.
  - arXiv: 2511.22606

Research Interests

Medical Image Processing, Medical Image Segmentation, Deep Learning for Radiology, Computer Vision,Uncertainty Quantification in Medical AI, Explainable AI (XAI) for Healthcare, AI-based Clinical Decision Support, Semantic Segmentation

Technical Skills

Languages: Python, C/C++, JavaScript, HTML + CSS, Dart, Java
Frameworks & Libraries: TensorFlow, Keras, PyTorch, Flutter, ReactJs, ASP.NET Core MVC, Bootstrap
ML/DL & CV Tools: OpenCV, NumPy, scikit-learn, matplotlib, CBAM, HiResCAM, EfficientNetB3, ResNet50, U-Net
Databases: MySQL, SQL Server, Firebase, MongoDB
Dev Tools: Git, GitHub, VS Code, XAMPP, phpMyAdmin, SSMS, .NET 8 SDK
Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Database Management System, Operating Systems, Software Engineering
Soft Skills: Research-driven Problem Solving, Self-learning, Adaptability, Presentation